

Why does therapeutic development take so long? the Philadelphia chromosomes and protein Kinases: Gleevec (part 2)

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Case 2: The Philadelphia chromosome and protein kinases: Gleevec

In the first installment of “Traces”, I explored the case of inhibitors of the mutated form of the Ras protein that took approximately four decades to move from being identified as a drug target to the approval of the first anti-cancer drugs. This week I discuss Gleevec, one of the first huge successes of targeting a protein encoded by genes that are specifically in cancer which moved through this process in less than a decade. The papers describing the clinical trials were breathtaking. I was teaching first year medical students at Johns Hopkins School of Medicine and included these papers as soon as they appeared. I literally cried when I first read them since 100% of the patients showed rapid improvement with minor side effects. This is almost unheard of in clinical trials. This ushered in a new era of anti-cancer drugs although other cancers proved to be more challenging, in large part because of their genetic heterogeneity.

In the discussion of mutated Ras inhibitors, the fact that cancer has its roots in genetic alterations was key. Evidence supporting this concept had been obtained prior to the evolution of modern tools for molecular cloning and DNA sequencing. Chronic myelogenous leukemia (CML) is a cancer of the white blood cells. Visualization of chromosomes can sometimes reveal substantial genetic rearrangements such as

translocations where two chromosomes have exchanged genetic material. The first of these was characterized in 1960 and was called the “Philadelphia chromosome”, in recognition of the location of the two institutions where researchers who first described it worked. As demonstrated in 1973, this translocation involves chromosomes 9 and 22 which have swapped parts of their longer arms. Remarkably more than 90% of all CML cases show the presence of this translation. Further examination of these translocations with modern tools in the early 1980s revealed that they generate a fused gene with the *BCR* (“breakpoint control region”) gene from chromosome 22 fused to the *ABL* gene from chromosome 9. The *ABL* gene encodes Abelson kinase, a protein kinase playing a role in controlling cell growth and division that is normally tightly regulated. But in the context of the translocated control region, it is continuously expressed. As with mutated forms of Ras, this yields cells with a control device stuck in the “ON” position.

Because this genetic change is so common in CML, ABL kinase was perceived as an attractive drug target. Protein kinases utilize adenosine triphosphate (ATP) to transfer phosphoryl groups to other proteins, and the phosphorylated protein products often have important roles in signaling pathways. An obvious strategy for developing drugs that target ABL kinase would be to find molecules that would compete with ATP and prevent these signaling processes from occurring. However, this reveals a huge challenge. ATP is utilized by many hundreds of enzymes within our cells, and it seemed very likely that finding compounds that would block ABL kinase without affecting many other enzymes would be next to impossible. But hematologist Brian Drucker developed a collaboration with a pharmaceutical company to hunt for such compounds and found one (STI-571) that worked well in cultured cells in 1993. Despite this, management of the company was reluctant to proceed further because of the high risk of side effects and toxicity related to inhibiting ATP-dependent enzymes. However, the company did proceed with clinical trials and the paper describing the results of the Phase 1 trial

revealed beautiful results. Eighty-three CML patients for which other therapies had failed were treated with the drug at a range of doses. The compound was, in general, well tolerated. Remarkably, all 54 patients receiving the highest tested dose showed substantial hematological responses, readily measurable from white blood cell counts, within less than a month, with limited side effects.

This drug, imatinib (Gleevec) was the first protein kinase inhibitor to be approved by the FDA, in 2001. The choice of CML was, in many ways, a best-case scenario since almost all patients carry the same genetic change and the results of treatment were readily monitored in a relatively non-invasive way. In contrast, most other cancers are genetically quite heterogeneous and tracking tumor progression can be more challenging and invasive with imaging studies and, in some cases, biopsies. Nonetheless, nearly 100 selective protein kinase inhibitor small molecules have now been FDA approved for treatment of different types of cancer. This is remarkable since, prior to the success with imatinib such selective kinase inhibition was thought to be impossible or, at least, improbable.



